

Final project report: Generating Faulty Math and Physics Questions Using In-Context Learning Prompts to fool LLM

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Abstract

This report explores the use of in-context learning (ICL) prompts to generate faulty mathematics and physics questions aimed at evaluating the reasoning capabilities of large language models (LLMs). By introducing subtle logical inconsistencies, these questions test the models' ability to detect and address flaws, providing insights into their compositional generalization skills.

1 Introduction

Large language models have demonstrated significant advancements in understanding and generating human-like text. However, their ability to perform complex reasoning, especially when faced with subtle logical inconsistencies, remains an area of active research. This study employs in-context learning prompts to create faulty math and physics questions, assessing LLMs' proficiency in identifying and reasoning through these challenges.

2 Characterization of Faulty Questions

The faulty questions crafted for this study are designed to expose specific vulnerabilities in large language models (LLMs). Each question introduces a subtle inconsistency that appears superficially valid, challenging the model's ability to integrate mathematical and physical principles into its reasoning. These questions are categorized into the following types:

2.1 Numerical Rule/surface level Violations

Questions exploit LLMs' potential lack of attention to fundamental numerical principles, such as the triangle inequality theorem or angle sum rules.

- **Example:** A triangle has sides of lengths 2 cm, 2 cm, and 5 cm. How long is the total sum of triangle.
- **Fault:** The side lengths violate the triangle inequality rule, making it impossible to form a triangle.
- **LLM Response Pattern:** The model often confirms validity, failing to check the numerical constraint.

2.2 Unrealistic Unit Conversions

These questions test the LLM’s ability to perform proper unit conversions and recognize implausible results.

- **Example:** A car travels 100 km in 2 hours. What is its average speed in km/s?
- **Fault:** Incorrect unit conversion leads to an unrealistic speed, which violates physical plausibility.
- **LLM Response Pattern:** The model performs the calculation without questioning the implausible output.

2.3 Logical Inconsistencies in Context

Logical inconsistencies involve altering contextual details to create conflicts with fundamental laws or established knowledge.

- **Example:** A pendulum oscillates at a frequency of 10 Hz with a length of 1 meter. Calculate the gravitational acceleration.
- **Fault:** The stated frequency is physically impossible for the given length, contradicting the pendulum’s natural frequency formula.
- **LLM Response Pattern:** The model computes a value without recognizing the incompatibility of the given parameters.

2.4 Real-World Plausibility Errors

Questions are mathematically or physically implausible but superficially reasonable.

- **Example:** A family has 2.5 children. How many children are boys?
- **Fault:** It is impossible for a family to have a fractional number of children.
- **LLM Response Pattern:** The model processes the question without flagging the logical inconsistency.

2.5 Conceptual Misunderstandings

These questions are designed to confuse models by exploiting closely related terms or concepts, such as speed and velocity or mass and weight.

- **Example:** A ball is dropped on the Moon where the gravitational acceleration is 9.8 m/s^2 . How long does it take to hit the ground from 10 meters?
- **Fault:** The Moon’s gravitational acceleration is incorrectly stated as 9.8 m/s^2 , which is Earth’s value.
- **LLM Response Pattern:** The model performs calculations based on the incorrect premise without recognizing the faulty assumption.

2.6 Surface-Level Plausibility Traps

These questions rely on phrasing that mimics valid queries but embed errors in logic or physics that require deeper reasoning to detect.

- **Example:** A spaceship traveling at $0.8c$ emits a light beam forward. The speed of the light relative to the spaceship is $1.25c$.
- **Fault:** This violates the invariance of the speed of light, contradicting Einstein’s theory of relativity.
- **LLM Response Pattern:** The model incorrectly confirms the statement, failing to apply relativistic principles.

2.7 Summary of Fault Types

Each question type targets a distinct reasoning deficiency in LLMs, ranging from mathematical rule violations to real-world plausibility errors. By categorizing these flaws, in future those datasets can systematically evaluate the models’ reasoning capabilities and identifies areas for improvement.

3 Methodology

This project leverages the MathTrap dataset and in-context learning (ICL) techniques to analyze logical loopholes in mathematical reasoning and apply them to generate faulty questions in both math and physics domains. The approach involves extracting common fault patterns, extending these patterns to physics, and evaluating the generated questions.

3.1 Analyzing Logical Loopholes in MathTrap

The MathTrap dataset consists of mathematical problems designed to exploit reasoning deficiencies in large language models (LLMs). Our analysis includes:

1. **Fault Categorization:** Group questions based on their logical flaws, such as:
 - Numerical rule violations (e.g., triangle inequality errors).
 - Unrealistic assumptions (e.g., fractional children in families).
 - Logical contradictions (e.g., incorrect angle sums).
2. **Pattern Identification:** Identify recurring fault patterns across the dataset, noting how these flaws are embedded in superficially plausible scenarios.
3. **Generalization Framework:** Develop a framework to extend these fault patterns into other domains, such as physics.

3.2 In-Context Learning Framework

In-context learning is used to extend fault patterns identified in MathTrap to generate new faulty questions. The ICL framework includes:

1. **Instruction:** The LLM is instructed to analyze examples of faulty questions and generate similar ones by embedding logical flaws.
2. **Examples:** Faulty questions from MathTrap are included as demonstrations, with annotations specifying the fault and reasoning.
3. **Prompt Template:** The ICL prompt is structured as follows:

Instruction:

You are tasked with analyzing and generating faulty questions in mathematics and physics. Faulty questions contain subtle logical flaws that make them invalid despite appearing plausible. Study the examples below and then generate similar questions.

Examples:

1. Question: A triangle has sides of lengths 2 cm, 2 cm, and 5 cm. Is this a valid triangle?
Reason: The side lengths violate the triangle inequality rule.
Expected Response: This triangle is invalid due to the triangle inequality rule.
2. Question: The interior angles of a quadrilateral are 90° , 100° , 120° , and 160° . Is this possible?
Reason: The sum of the angles exceeds the required 360° for a quadrilateral.
Expected Response: This question is invalid due to the incorrect angle sum.

Your Task:

Generate two new questions|one in mathematics and one in physics|by embedding logical flaws. For each question, provide:

- The faulty question.
- The reason for the fault.
- The expected response.

4. **Task Prompting:** After the examples, the LLM is prompted to generate faulty questions in math and physics using the patterns identified.

3.3 Extending to the Physics Domain

The logical fault patterns identified in the MathTrap dataset are adapted to physics by:

1. **Mapping Patterns:** Translate mathematical flaws into equivalent physics scenarios. For instance:
 - Triangle inequality violations are adapted to physical constraints, such as invalid velocity combinations.

- Logical inconsistencies (e.g., incorrect angle sums) are adapted to violations of physical laws, such as energy conservation.
2. **Domain-Specific Faults:** Introduce new inconsistencies specific to physics, such as incorrect unit conversions or implausible gravitational assumptions.
 3. **ICL Application:** Use the same ICL framework and prompt template to guide the LLM in generating new faulty physics questions.

3.3.1 Prompt Template

Instruction:

You are tasked with analyzing faulty questions from the mathematical domain and adapting their logical flaws to generate analogous faulty questions in the physics domain. Faulty questions contain subtle logical inconsistencies that make them invalid despite appearing plausible. Study the examples below and follow the patterns to create new faulty questions in the physics field.

Examples of MathTrap Questions:

1. Question: A triangle has sides of lengths 2 cm, 2 cm, and 5 cm. Is this a valid triangle?
Reason: The side lengths violate the triangle inequality rule, making it impossible to form a triangle.
Expected Response: This triangle is invalid due to the triangle inequality rule.
2. Question: The interior angles of a quadrilateral are 90° , 100° , 120° , and 160° . Is this possible?
Reason: The sum of the angles exceeds the required 360° for a quadrilateral.
Expected Response: This question is invalid due to the incorrect angle sum.
3. Question: A family has an average of 2.5 children. How many children are boys?
Reason: It is impossible for a family to have a fractional number of children, making the scenario logically flawed.
Expected Response: This question is invalid due to the implausible premise of fractional children.

Adaptation to Physics:

Using the logic of the above examples, apply similar patterns to create faulty questions in physics. For instance:

- Numerical rule violations in math (e.g., triangle inequality) can translate to violations of physical constraints (e.g., speed, force, or energy limits).

- Unrealistic values in math (e.g., fractional children) can map to implausible physical parameters (e.g., gravitational acceleration, velocity).
- Logical contradictions in math (e.g., angle sum errors) can correspond to violations of physical laws (e.g., conservation of energy, relativity).

Examples of Physics Questions:

1. Question: A spaceship traveling at $0.8c$ emits a light beam forward. The speed of the light relative to the spaceship is $1.8c$. True or false?
Reason: This violates the invariance of the speed of light, as per Einstein's theory of relativity.
Expected Response: This question is invalid because the speed of light is constant in all reference frames.
2. Question: A pendulum oscillates at a frequency of 10 Hz with a length of 1 meter. Calculate the gravitational acceleration.
Reason: The stated frequency is physically impossible for a pendulum of this length under Earth's gravity, violating the pendulum's natural frequency formula.
Expected Response: This question is invalid due to the implausible frequency for the given length.

Your Task:

Generate two faulty physics questions based on the MathTrap patterns.

For each question, provide:

- The faulty question.
- The reason for the fault.
- The expected response.

3.4 Evaluation Process

The evaluation process includes:

1. **Fault Detection:** Presenting the generated questions to the LLM to assess its ability to recognize and flag logical flaws.
2. **Reasoning Validation:** Tasking the LLM to solve faulty questions and evaluate whether it identifies the embedded flaws.

3.5 Summary

By combining the analytical insights from MathTrap with the generative capabilities of in-context learning, this project successfully creates new around 100 faulty questions in both math and physics that evaluated by human and powerful LLM such as Qwen and chatgpt. This methodology provides a robust framework for testing LLM reasoning across domains.

4 Possible Defenses Against Logical Errors and faulty questions in LLMs

In this section we will discuss some possible strategy that can possible help LLM recognize faulty questions. Below, we outline potential defenses to mitigate logical errors, focusing on techniques such as Chain-of-Thought (CoT) prompting and using LLMs as self-evaluative judges.

4.1 Chain-of-Thought (CoT) Reasoning

Chain-of-Thought (CoT) prompting is a technique where the LLM is explicitly guided to break down its reasoning process into smaller, logical steps. By encouraging a step-by-step approach, CoT reduces the likelihood of the model making surface-level errors.

Implementation of CoT

To resolve logical errors in faulty questions, CoT prompts the LLM to:

1. Decompose the problem into sequential reasoning steps.
2. Validate each step against mathematical or physical principles.
3. Arrive at a conclusion that considers all intermediate validations.

Example: For the faulty question, "A triangle has sides of lengths 2 cm, 2 cm, and 5 cm. Is this a valid triangle?"

Prompt:

Let's solve this step by step:

1. Recall the triangle inequality rule: the sum of any two sides must be greater than the third side.
2. Check: $2 + 2 > 5$. Is this true? No, because 4 is not greater than 5.
3. Conclusion: The triangle inequality is violated, so this is not a valid triangle.

This structured approach helps the LLM validate logical premises before reaching a conclusion.

4.2 LLMs as Judges

Another effective defense is using an LLM in a self-evaluative or adversarial role. By prompting the model to critique its own responses or those generated by another LLM, it can identify inconsistencies or errors more effectively.

Implementation of LLMs as Judges

To apply this method:

1. Generate an initial response to a faulty question.

2. Provide the response to the LLM with an additional prompt asking it to critique the reasoning.
3. Iterate this process until the LLM identifies and corrects any logical errors.

Example: For the faulty physics question, "A pendulum oscillates at a frequency of 10 Hz with a length of 1 meter. Calculate the gravitational acceleration."

Initial Response:

$g = (2\pi f)^2 * L$
 $g = (2\pi * 10)^2 * 1 = 3947.84 \text{ m/s}^2$.

Judge Prompt:

Critique the reasoning above. Does the result align with realistic gravitational acceleration values on Earth? Identify any flaws in the assumptions or calculations.

This encourages the model to flag the unrealistic frequency assumption and revise its answer.

4.3 Combining CoT and LLMs as Judges

The combination of CoT reasoning and LLMs as judges creates a robust framework for logical validation:

1. Use CoT prompting to guide the model through its reasoning process.
2. Apply the "LLM as a judge" approach to evaluate the CoT-generated response.
3. Iterate the process until the response satisfies logical and contextual correctness.

4.4 Future Directions for Defenses

Additional strategies to address logical errors could include:

- **Fine-Tuning with Faulty Datasets:** Incorporating datasets like MathTrap into LLM training pipelines to expose models to flawed reasoning during pretraining.
- **Prompt Engineering for Robustness:** Designing prompts that explicitly request models to validate premises before making calculations or assumptions.

5 Conclusion

This project demonstrates that using ICL prompts to generate faulty questions is an effective method for evaluating the reasoning capabilities of LLMs. The results underscore the need for continued research to enhance the compositional generalization skills of these models, aligning with ongoing efforts in the field.

References

- Zhao, J., Tong, J., Mou, Y., Zhang, M., Zhang, Q., Huang, X. (2024). Exploring the Compositional Deficiency of Large Language Models in Mathematical Reasoning. *arXiv preprint arXiv:2405.06680*. <https://arxiv.org/abs/2405.06680>